**Unit:** Unit 0: Foundations of Algebra **Name:** ____________
**Topic:** GCF LCM **Date:** __________

## Advanced (Habanero)

**Q1.** What is the GCF of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

**Q2.** What is the LCM of 9 and 12?
(A) 12
(B) 18
(C) 24
(D) 36
(E) 48

**Q3.** What is the GCF of 24 and 30 using prime factorization?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

**Q4.** A bookshelf has 12 shelves, and each shelf can hold 8 books. What is the LCM of 12 and 8?
(A) 12
(B) 16
(C) 24
(D) 32
(E) 48

**Q5.** What is the GCF of 15 and 20?
(A) 1
(B) 3
(C) 5
(D) 10
(E) 15

**Q6.** A group of friends want to share some candy equally. If they have 18 pieces of candy and there are 3 friends, what is the LCM of 18 and 3?
(A) 6
(B) 9
(C) 12
(D) 18
(E) 27

**Q7.** What is the GCF of 8 and 12 using factors?
(A) 1
(B) 2
(C) 4
(D) 8
(E) 12

**Q8.** A teacher has 15 boxes of pencils, and each box contains 12 pencils. What is the LCM of 15 and 12?
(A) 30
(B) 36
(C) 48
(D) 60
(E) 72

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**Open Ended Questions (FRQ)**

**Q9.** Tom has 18 pounds of apples and 12 pounds of oranges. He wants to put an equal number of pounds of each fruit into boxes. What is the largest number of pounds of each fruit that Tom can put into each box?

**Q10.** A factory produces 12-packs and 15-packs of batteries. What is the least number of packs the factory needs to produce in order to have the same number of batteries in each type of pack? Show your work and explain your reasoning.

**Chef's Tips**



- Remind students to use prime factorization when finding GCF or LCM
- Encourage students to draw diagrams or use visual aids to help solve problems
- Emphasize the importance of checking work and explaining reasoning